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DEALER BEST PRACTICE SERIES

Forecasting CRC Component Rebuild Volume and Capacity

Application | Maintenance | Site Management | **Component Rebuild** | Component Life Management | Safety | MARC Management

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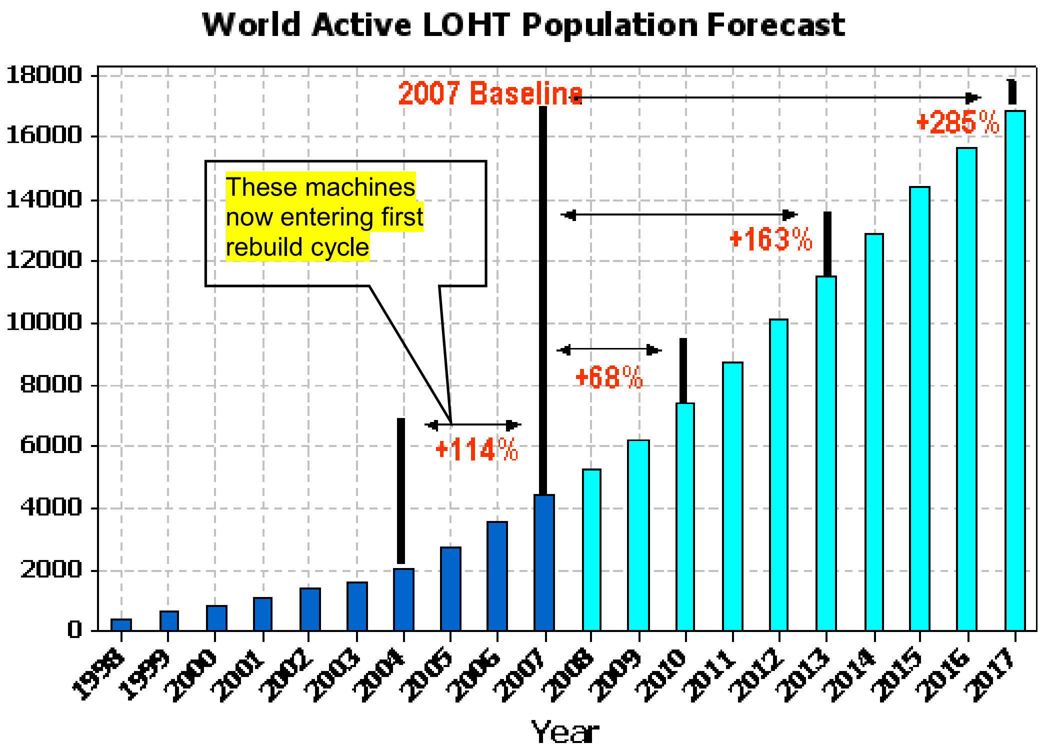
1.0 Introduction

Forecasting future component rebuild volumes enables the dealer to make plans to manage future customer requirements and to forecast future revenues. CRC component rebuild volume from mining customers is directly tied to the active mining machine populations and changes to that population.
Over the past three years, the worldwide population of Caterpillar Large Off Highway Trucks (785 and up) has doubled. Forecasts indicate that the active population of machines will double again in the next three to four years.
Populations for other mining machine models for mining markets are showing similar growth forecasts.
This dramatic growth of active mining machine populations will impact future CRC component rebuild volumes. Future customer component rebuild requirements may exceed current CRC capacity and create a need for expansion and/or efficiency improvement.
This Best Practice is a tool to forecast future component rebuild volumes, to assess current capacity, and to develop a “Five Year Plan” to manage the growth in the Mining business.
This Best Practice is a Caterpillar / Dealer joint project from APD.



- Includes 785 – 797 only
- Assumes 12 year machine life
- Forecast sales based on unconstrained supply

2.0 Best Practice Description

This Best Practice is a process to assess current active mining machine population in a dealer territory and to forecast the growth in machine population. With that data, a dealer can estimate the expected increase in major component rebuilds the machine population growth will generate. The expected future component rebuild needs can then be balanced against the CRC maximum capacity given current processes and assets. Plans can then be made to manage the change.

3.0 Implementation Steps

- The first step is to determine which mining machine models families create the majority of the CRC component rebuilds. The chart below is a dealer analysis of CRC rebuilds by machine model family (2007 data).

DESCRIPTION	Total	%
TRUCK	217	44%
TRACK TYPE TRACTOR	89	18%
WHEEL LOADER	56	11%
MOTOR GRADER	45	9%
WHEEL DOZER	33	7%
EXCAVATOR	13	3%
WATER TANK	8	2%

Off High Trucks clearly have the largest impact, however, the other machine models combined also generate significant rebuild volume.

- The next step is to gather machine population data for the machines models generating most of the CRC rebuild activity. Refer to the OHT population chart in section one above. To develop the chart, LOHT machine sales data was collected for the past twelve years and forecast sales data for the next 10 years. The average life of a LOHT was assumed to be 12 years.
  To calculate the current "active" population, add together all the sales for the past twelve years. To calculate future population, add the forecast sales and subtract the machines that are more than 12 years old for each year.
  It is helpful to create a graph similar to the graph in section one.
  This process should be repeated for each major family of mining machines.
  - Large Off Highway Trucks
  - Track Type Tractors
  - Wheel Loaders
  - Motor Graders
  - Wheel Dozers

- The next step is to calculate the percentage of active machine population increase from each year to the next year. The percent growth of machine population can be used to predict expected growth in CRC component rebuilds. The "percent" change in population should relate directly to a "percent" change in CRC component rebuilds.

  Example Forecast:

LOHT Population (785 - Up)			
Year	**Sales**	**Accumulated Population ***	**Percent Change**
1998	18	18	N/A
1999	17	35	94%
2000	19	54	54%
2001	22	76	41%
2002	26	102	34%
2003	29	131	28%
2004	31	162	24%
2005	31	193	19%
2006	41	234	21%
2007	62	296	26%
2008	81	377	27%
Forecast			
2009	112	489	30%
2010	129	600	23%
2011	147	712	19%
2012	159	817	15%
2013	175	916	12%
2014	195	1009	10%

* Assumes 12 year machine life

Example:

1. Sales history data entered in table.
2. Sales forecast data entered in table.
3. Active population calculated assuming 12-year machine life.
4. Year to year change calculated.

Planned machine component rebuilds occur two years after a new machine is delivered. Today's (2008) planned rebuilds in the shop are from machines sold prior to 2006. The rebuilds next year (2009) will be for machines sold before 2007. The rebuild forecast for next year is then based on the population change from 2006 to 2007.

LOHT Population (785 - Up)				
Year	Sales	Accumulated Population	Percent Change	Forecast Rebuilds
1998	18	18	N/A	
1999	17	35	94%	
2000	19	54	54%	
2001	22	76	41%	
2002	26	102	34%	
2003	29	131	28%	
2004	31	162	24%	
2005	31	193	19%	
2006	41	234	21%	
2007	62	296	26%	200
2008	81	377	27%	242
Forecast				
2009	112	489	30%	307
2010	129	600	23%	391
2011	147	712	19%	507
2012	159	817	15%	622
2013	175	916	12%	738
2014	195	1009	10%	847

2007 Actual Rebuilds

Example:

Dealer rebuilt 200 OHT components in the CRC in 2007.

What is the forecast for 2008?

- Rebuilds in 2008 are on machines sold prior to 2006.
- Growth in 2006 was 21%
- 2008 Forecast;
  200 +21% = 242

What is forecast for 2009?

- Based on pre-2007 machines
- 2007 growth 26%
- 2009 Forecast:
  242 + 26% = 307

- Assumes 12 year machine life

Note: In the example above, with only a 20-30% annual growth in machine population, the CRC component rebuilds will go from 242 in 2008 to 507 in 2011. The expected CRC component rebuild requirements double in the next three years.

A blank Excel spreadsheet is attached to the Best Practice. Enter machine sales history, sales forecast, and your 2007 CRC component rebuild volume, and the spreadsheet will calculate expected forecast volumes.

Assumptions:
This process for estimating future CRC component rebuild volumes is based on a number of important assumptions. These assumptions should be carefully considered for validity.

- Assumes a 12-year average machine life. If life is longer, the impact is higher active machine populations. Shorter life lowers active machine populations.
- Assumes machine sales forecast is reasonably accurate. Commodity prices maintain the growth in Mining. Mines that may be near the end of their productive life need to be considered. New mines being opened need to be considered in the machine sales forecast.
- Assumes that customers performing all or a part of their own component rebuilds will not change policy. There is some risk that mines may not be able to continue their own rebuilds with expanded fleets and personnel shortages.
- Assumes that dealer and customers policy on use of Reman components will not change. Using Reman major components displaces component rebuilds, so a change in policies changes CRC rebuild volumes.
- Assumes that the balance between planned and unplanned component rebuilds will not change. Level of unplanned component rebuilds (failures) will not increase or decrease CRC rebuild volume.
- Assumes CRC rebuilds are proportional to active population of machines.

Using the forecast: CRC Planning

Once a long range forecast for CRC component rebuild volume is determined, that forecast needs to be compared to the present CRC capacity given manpower, shifts, facility, tooling, and processes.
Where forecasted component rebuild requirements within the next few years exceed present CRC capacity, it is appropriate to develop a plan to manage the future component rebuild requirements.

Options for Increasing CRC Capacity

- Improve Component Turnaround Time (efficiency)
  - Eliminate bottlenecks
  - Eliminate waste (lean processes)
  - Optimize asset utilization (planning and scheduling)
  - Streamline processes
- Increase Resources
  - Add / retain technicians
  - Add / expand shifts
  - Add facility and work bay space
  - Add / modernize tooling
- Utilize Reman Major Components for Peak Shaving
- Utilize Reman subcomponents in lieu of salvage
- Increase Parts and Spare Component Inventories.

4.0 Benefits

A long-range component rebuild forecast allows a dealer CRC to:

- Plan to meet mining customer future component rebuild needs.
- Utilize dealer assets effectively.

5.0 Resources Required

- Management time to collect data, analyze, and review findings

6.0 Supporting Attachments / References

Components rebuild forecast spreadsheet.
OHT sales forecasts by region.

7.0 Related Best Practices

- Improving CRC Component Turnaround Time
- CRC Planning & Scheduling Systems
- Labor Efficiency Lowers CRC Costs

Editor's note: Although the mining business is becoming ever more cyclical, the principals of this best Practice during a projected growth cycle are still relevant.

8.0 Acknowledgements

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